

DS Unit 3 — Trees & Binary Search Trees

Complete Notes

Data Structures (DS) · Unit 3 · Trees & Binary Search Trees · 2025

Overview

These notes cover **Trees & Binary Search Trees**. Core data structures and the algorithms built on them — arrays, lists, stacks, trees, graphs, sorting and hashing. Each section states the core idea, a worked intuition and the points examiners look for.

1. Binary Tree Traversals

Inorder, preorder, postorder and level-order traversals visit every node in a defined order.

Why it matters: Binary Tree Traversals appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of binary tree traversals in one line.
- A small example showing binary tree traversals in practice.
- Common mistakes students make and how to avoid them.

2. Binary Search Trees

In a BST every left descendant is smaller and every right descendant larger, enabling $O(h)$ search.

Why it matters: Binary Search Trees appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of binary search trees in one line.
- A small example showing binary search trees in practice.
- Common mistakes students make and how to avoid them.

3. AVL Trees

AVL trees keep balance factors in $\{-1, 0, 1\}$ using LL, RR, LR and RL rotations.

Why it matters: AVL Trees appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of avl trees in one line.
- A small example showing avl trees in practice.
- Common mistakes students make and how to avoid them.

4. Heaps

A binary heap is a complete tree satisfying the heap property, used for heapsort and priority queues.

Why it matters: Heaps appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of heaps in one line.
- A small example showing heaps in practice.
- Common mistakes students make and how to avoid them.

5. B-Trees

B-trees are balanced multiway search trees designed to minimise disk accesses.

Why it matters: B-Trees appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of b-trees in one line.
- A small example showing b-trees in practice.
- Common mistakes students make and how to avoid them.

6. Threaded Binary Trees

Threads replace null pointers with links to inorder predecessors or successors.

Why it matters: Threaded Binary Trees appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of threaded binary trees in one line.
- A small example showing threaded binary trees in practice.
- Common mistakes students make and how to avoid them.

Quick revision

[Binary Tree Traversals](#) · [Binary Search Trees](#) · [AVL Trees](#) · [Heaps](#) · [B-Trees](#) · [Threaded Binary Trees](#)